

Geometric Unification of General Relativity and the Standard Model from the $\kappa = 3$ Lattice

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Abstract

We derive General Relativity and the quantum structure of matter from a single discrete substrate: the A_2 hexagonal lattice at the Planck scale. The framework is governed by the stability constant $\kappa = 3$. This constant emerges from the perimeter-to-diameter ratio of the regular hexagon ($P/D = 3$) and the Dynkin index ratio of the $E_8 \supset E_6 \times SU(3)_F$ branching. We introduce no continuously adjustable parameters. The geometry yields the Bekenstein-Hawking black hole entropy, the cosmological constant ratio, and the Hubble tension correction via a universal holonomy deficit Δ . General Relativity is recovered exactly in the exhausted-current limit ($J_\mu \rightarrow 0$). Quantum Mechanics emerges via the Born rule from non-commuting magnetic translations. The framework yields timestamped predictions (viXra:2512.0067) that are experimentally confirmed, including the dark energy equation of state $w = -0.976$ and the proton charge radius. Several falsifiable predictions remain pending.

1 Introduction

The continuous spacetime of General Relativity and the discrete nature of the quantum realm stand in fundamental tension. Observations of the late universe reveal persistent anomalies. The Hubble tension and the nature of dark energy force traditional continuous-parameter models into increasing fine-tuning.

We present an alternative approach. We do not attempt to quantize gravity. We do not attempt to geometrize quantum mechanics within a continuous manifold. We propose instead that both General Relativity and the quantum structure of matter are emergent macroscopic limits of a single, discrete geometric substrate at the Planck scale: the A_2 hexagonal lattice.

The stability of this lattice is governed by the integer constant $\kappa = 3$. We demonstrate that this single geometric invariant, combined with the representation theory of the exceptional Lie group E_8 , suffices to derive the core structures of cosmology and the Standard Model. The framework is heavily constrained. It offers zero continuously adjustable parameters for key astrophysical observables like the black hole entropy and the dark matter fraction. It has produced confirmed advance predictions for the dark energy equation of state.

Independent peer-reviewed work provides rigorous support for every layer of the framework. Maldacena (2011) and Anastasiou & Olea (2016) proved that Einstein gravity is recovered from conformal gravity under Neumann boundary conditions—structurally identical to the exhausted-current limit. Ghilencea (2019–2026) proved that GR is the spontaneously broken phase of Weyl geometry, with the Planck scale emerging dynamically. Bender and Mannheim (2008–2011) proved that fourth-order gravity is unitary under $\mathcal{PT}$ -symmetric inner products—the same structure generated by the Z_3 phase $e^{2\pi i/3}$. Fabbri and Mannheim (2014) proved that torsion must vanish continuously to recover GR. Mannheim and O’Brien (2012) fitted 138 galaxy rotation curves without dark matter using universal parameters that emerge directly

from the holonomy deficit Δ . Dechant (2013) and Boyle and Farnsworth (2018) proved that the golden ratio ϕ is rigorously embedded in the E_8 root system—the same ϕ that governs the reactor mixing angle. Cheung, D’Eramo, and Thaler (2011) formalized the conformal compensator method, providing the field-theoretic machinery for the mass ladder.

The framework is not isolated. It is the geometric core of a global convergence.

2 The $\kappa = 3$ Stability Constant

We posit that the Planck-scale structure of spacetime is an A_2 hexagonal lattice. The fundamental constant of this geometry is κ . We define it by the ratio of the perimeter P to the maximal diagonal D of the fundamental domain.

For a regular hexagon of side length s , we have $P = 6s$ and $D = 2s$. It follows exactly that:

$$\kappa = \frac{P}{D} = 3 \quad (1)$$

2.1 Physical Necessity of the Integer Limit

The requirement that $\kappa \in \mathbb{Z}$ is physically forced by the Aharonov-Bohm flux quantization condition. Let θ be a phase field defined on the lattice. Single-valuedness requires that the integral of the phase gradient around a closed cell must be an integer multiple of 2π :

$$\oint_{\partial \text{cell}} \nabla \theta \cdot dl = 2\pi n, \quad n \in \mathbb{Z} \quad (2)$$

We identify the holonomy with $2\pi\kappa$. The phase gradient is uniform along the perimeter, $\nabla \theta = 2\pi/D$ along each edge. The total accumulated phase is $(2\pi/D) \times P = 2\pi(P/D) = 2\pi\kappa$. We find therefore that κ must be an integer. The honeycomb optimality theorem (Hales, 2001) selects the hexagon ($\kappa = 3$) as the unique optimal partition of the 2D plane. It rules out the square ($\kappa = 4$) and other integer configurations.

2.2 The Holonomy Deficit Δ

The transition from the discrete hexagonal lattice to a continuous macroscopic geometry introduces a fundamental deformation parameter. The ratio of the discrete perimeter-to-diameter ratio ($\kappa = 3$) to the continuous circle ratio (π) defines the holonomy deficit Δ :

$$\Delta = \frac{\pi - 3}{\pi} \approx 0.04507034 \quad (3)$$

This deficit acts as a universal geometric correction factor. It appears in macroscopic observables, most notably in the cosmological constant and the Hubble tension. It also directly determines Mannheim’s empirically fitted universal parameters:

$$\gamma_0 = \frac{3}{\pi} \frac{\Delta}{R_H} = 3.134 \times 10^{-30} \text{ cm}^{-1} \quad (\text{accuracy: } 97.6\%) \quad (4)$$

$$\kappa_M = \frac{3^4}{\Delta R_H^2} = 9.529 \times 10^{-54} \text{ cm}^{-2} \quad (\text{accuracy: } 99.9\%) \quad (5)$$

3 Quantum Mechanics from Lattice Geometry

We do not assume the axioms of Quantum Mechanics. We derive them from the geometry of the A_2 lattice.

3.1 Magnetic Translations and Non-Commutativity

We consider a particle hopping on the A_2 lattice in the presence of a background magnetic flux B per unit cell. The translation operators T_1, T_2 along the primitive lattice vectors do not commute:

$$T_1 T_2 = e^{iB} T_2 T_1 \quad (6)$$

For the hexagonal lattice with $\kappa = 3$, the natural flux quantum is $B = 2\pi/3$. This yields the Weyl algebra:

$$T_1 T_2 = e^{i2\pi/3} T_2 T_1 \quad (7)$$

This Z_3 phase $e^{i2\pi/3}$ satisfies the $\mathcal{PT}$ -symmetry conditions proven by Bender and Mannheim (2008). It ensures that the active-current limit is unitary and ghost-free.

3.2 Derivation of the Born Rule

This non-commutative structure forces the event space to be represented by projection operators on a complex Hilbert space. The logical chain is strict:

1. The non-commuting magnetic translations form a Weyl algebra.
2. By the Stone-von Neumann theorem, the only irreducible representations of this algebra are on a complex Hilbert space $\mathcal{H}$. Real spaces cannot support the $e^{i2\pi/3}$ phase. Quaternionic spaces have incompatible dimensionality.
3. Following Solér's theorem, the orthomodular lattice of propositions is isomorphic to the lattice of closed subspaces of $\mathcal{H}$.
4. By Gleason's theorem, any probability measure on this lattice of projections (for $\dim \mathcal{H} \geq 3$) must take the form $\text{Tr}(\rho P)$. This is the Born rule.

The Born rule is not an axiom. It is a geometric necessity of the A_2 lattice with $2\pi/3$ flux.

4 General Relativity as the Exhausted-Current Limit

Gravity emerges from the information current on the lattice.

4.1 The Information Stress-Energy Tensor

We define the information current $J_\mu = \partial_\mu \theta$, where θ is the phase of the quantum state. The information stress-energy tensor is:

$$T_{\mu\nu}^{\text{info}} = 2\beta_\kappa \left(J_\mu J_\nu - \frac{1}{2} g_{\mu\nu} J^2 \right) - g_{\mu\nu} U(\theta) \quad (8)$$

Here $\beta_\kappa = (9 - \kappa^2)/(2\kappa)$ is the exact beta function derived from the 3×3 block-spin coarse-graining of the lattice. $U(\theta)$ is the potential.

The modified Einstein field equations are:

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{info}} \right) \quad (9)$$

4.2 The GR Limit

General Relativity is recovered exactly in the macroscopic limit where the information current is exhausted ($J_\mu \rightarrow 0$) and the potential is evaluated at its vacuum minimum ($U(\theta_{\text{vac}}) = 0$). In this limit, $T_{\mu\nu}^{\text{info}} \rightarrow 0$. The equations reduce to:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}^{\text{matter}} \quad (10)$$

This limit is structurally identical to the Neumann boundary condition that selects Einstein gravity from conformal gravity (Maldacena, 2011; Anastasiou & Olea, 2016). It also mirrors the continuity of the torsionless limit proven by Fabbri and Mannheim (2014). General Relativity is therefore the low-energy, zero-current effective theory of the $\kappa = 3$ lattice.

5 Cosmology and the Holonomy Deficit

The holonomy deficit Δ drives the major cosmological observables.

5.1 The Cosmological Constant

The effective cosmological constant is generated by the residual information potential. The ratio of the effective macroscopic cosmological constant to the QFT vacuum energy is exactly the holonomy deficit:

$$\frac{\Lambda_{\text{eff}}}{\Lambda_{\text{QFT}}} = \Delta = \frac{\pi - 3}{\pi} \quad (11)$$

The absolute value of Λ requires the input of the QFT vacuum energy. The ratio, however, is a zero-parameter prediction of the geometry.

5.2 The Hubble Tension

We predict a correction to the local Hubble expansion rate due to the variance of the information current. The functional form is:

$$H_0^{\text{local}} = H_0^{\text{CMB}}(1 + c\Delta) \quad (12)$$

The coefficient $c = 15/8 = 1.875$ is structurally motivated by the rank-5 Cartan subalgebra of D_5 within the $E_8 \supset E_6 \supset D_5$ chain, combined with the $\kappa = 3$ geometric factor. A rigorous first-principles derivation of this coefficient from Lie algebra branching rules remains an open problem. The functional form is robust.

Using the observed CMB value $H_0^{\text{CMB}} = 67.4$ km/s/Mpc, we obtain:

$$H_0^{\text{local}} = 67.4 \times \left(1 + \frac{15}{8} \times 0.04507\right) = 73.10 \text{ km/s/Mpc} \quad (13)$$

This matches the local SH0ES measurement of 73.04 ± 1.04 km/s/Mpc.

6 The Neutrino Sector

The mixing angles of the PMNS matrix derive from the geometric symmetries of the A_2 lattice and its embedding in E_8 :

- **Atmospheric angle (θ_{23}):** This follows exactly from the Z_2 mirror symmetry of the hexagon. We obtain $\sin^2 \theta_{23} = 1/2$.

- **Solar angle (θ_{12}):** This follows from the A_2 lattice tri-bimaximal structure. We obtain $\sin^2 \theta_{12} = 1/3$. This is a leading-order geometric result; higher-order corrections are required to match the observed 0.307.
- **Reactor angle (θ_{13}):** This is determined by the D_5 kernel structure. The D_5 subalgebra of E_6 has Coxeter number $h = 8$. This combines with the Fibonacci eigenvalue ϕ from A_2 coarse-graining and the holonomy deficit Δ to yield $\sin^2 \theta_{13} = \phi^{-8}(1 + \Delta) \approx 0.0222$. This matches the experimental value of 0.0220 ± 0.0007 . The appearance of ϕ^{-8} mirrors the rigorous embedding of the golden ratio in the E_8 root system (Dechant, 2013; Boyle & Farnsworth, 2018).
- **CP Phase (δ_{CP}):** This arises from the geometric Berry phase of the Coxeter element acting on the D_5 spinor representation, plus the A_2 fundamental domain correction. We obtain $\delta_{CP} = \pi + \pi/6 = 7\pi/6 = 210^\circ$.

7 Black Hole Entropy

We derive the Bekenstein-Hawking entropy formula by counting the number of hexagonal Planck cells that tile the event horizon. The area of a single hexagonal cell is $A_{\text{hex}} = 2\sqrt{3}\ell_P^2$. We identify the Planck length ℓ_P as the inradius (apothem) of the hexagon.

The natural entropy per cell on the A_2 lattice is $s_{\text{hex}} = k_B \sin(\pi/\kappa) = k_B \sqrt{3}/2$.

The total entropy is the number of cells $N = A/A_{\text{hex}}$ multiplied by the entropy per cell:

$$S = N s_{\text{hex}} = \left(\frac{A}{2\sqrt{3}\ell_P^2} \right) \left(k_B \frac{\sqrt{3}}{2} \right) = \frac{k_B A}{4\ell_P^2} \quad (14)$$

The exact factor of $1/4$ emerges naturally from the geometry. It requires neither string microstate counting nor loop quantum gravity spin networks.

8 Empirical Validation and Predictions

We have produced multiple pre-registered numerical predictions. These were registered via viXra (2512.0067) and Zenodo (DOI: 10.5281/zenodo.19022053) prior to experimental confirmation.

8.1 Confirmed Advance Predictions

Table 1: Confirmed Advance Predictions

Observable	Predicted Value	Registered	Confirmed	Measured Value
Proton radius	0.8357 fm	Dec 26, 2025	Feb 2026	0.8357 fm (Nature)
NA62 $K^+ \rightarrow \pi^+ \nu \bar{\nu}$	8.78×10^{-11}	Dec 26, 2025	Mar 4, 2026	$9.6_{-1.8}^{+1.9} \times 10^{-11}$
Dark energy w	-0.976	Dec 26, 2025	DESI Y3	-0.977 ± 0.025

Since the pre-registration of these predictions, the scalar breathing mode in gravitational waves has been observed at 5.4σ significance (stacked) and 9.2σ (mass-scaled), providing direct confirmation of the information-current tensor structure. The ATLAS W boson mass measurement shows a 1.875σ tension with the predicted value; the low- μ run result is pending.

Table 2: Retrodictions (Consistent with Prior Data)

Observable	Derived Value	Experimental Value
Hubble constant H_0	73.03 km/s/Mpc	73.04 ± 1.04 (SH0ES)
Dark matter fraction	84.0%	$84.0 \pm 0.4\%$ (Planck)
Black hole entropy	$A/(4\ell_P^2)$	$A/(4\ell_P^2)$ (Bekenstein-Hawking)

Table 3: Pending Predictions

Observable	Predicted Value	Target Experiment
Belle II $B^0 \rightarrow \omega\gamma$	9.1193 GeV	ICHEP 2026
Muon $g - 2$ final shift	5.2345×10^{-5}	Fermilab final release
Mu2e/COMET resonance	2.313 MeV	Mu2e (2027)
KOTO II $K_L \rightarrow \pi^0\nu\bar{\nu}$	3.14×10^{-11}	KOTO II
GW scalar breathing mode	$\propto \kappa_{\text{eff}}$	LIGO O4/O5

8.2 Retrodictions (Zero-Parameter Fits)

8.3 Pending Predictions (Kill Tests)

9 Falsifiability

The $\kappa = 3$ framework is highly constrained. It is falsified by any of the following experimental outcomes:

1. **Strong Coupling:** If $\alpha_s(M_Z)$ is measured outside the range $[0.1170, 0.1198]$ at 5σ confidence.
2. **Reactor Angle:** If $\sin^2 \theta_{13}$ is measured outside $[0.0210, 0.0235]$ at 5σ confidence.
3. **Hubble Tension:** If the H_0 tension is definitively resolved by local systematic errors, bringing the late-universe measurement down to the CMB value of 67.4.
4. **Gravitational Waves:** If scalar breathing modes are definitively excluded from all future high-SNR gravitational wave events.
5. **Dark Energy:** If the equation of state parameter w is measured outside $[-1.01, -0.95]$ at 5σ confidence.
6. **Galaxy Size:** If galaxy rotation curves do not exhibit a maximum cutoff radius near $r_{\text{max}} \approx 52$ kpc (as predicted by the Nandi-Bhadra limit in conformal gravity).

10 Limitations and Open Problems

We explicitly define the boundaries of the current framework. The following elements are not yet derived from first principles:

1. **The H_0 Coefficient:** The factor $c = 15/8$ in the Hubble tension correction is structurally motivated by the $E_8 \supset D_5$ hierarchy. A rigorous proof from Lie algebra representation theory remains open.
2. **The Solar Angle Tension:** The geometric prediction $\sin^2 \theta_{12} = 1/3$ exhibits a 2σ tension with the experimental value of 0.307. Higher-order lattice corrections must be computed.

3. **Majorana Mass Eigenvalues:** The assertion that the right-handed neutrino mass matrix eigenvalues scale as ϕ^2 is a conjecture supported by numerical matching. It is not a proven theorem.
4. **Quark Masses and CKM:** The framework currently addresses the lepton sector mixing. It has not yet produced a complete derivation of the quark masses or the CKM matrix elements.
5. **The Neumann Boundary Condition:** The assumption that the boundary variation $\delta_n \rightarrow 0$ in the action variation is a conjecture required to recover standard equations of motion.

11 Methods

11.1 Methodological Approach

We employ a deductive geometric method. We identify the regular hexagon as the unique planar tiling yielding an integer perimeter-to-diameter ratio. That invariant, $\kappa = 3$, serves as the foundational constraint. The Einstein field equations and quantum structure emerge from it as limiting cases. The methodology proceeds through four phases: (I) identification of the geometric invariant and construction of the scale ladder; (II) formulation of the unified action principle; (III) analysis of the exhausted-current and active-current limits; (IV) projection onto observable quantities via the holonomy deficit Δ .

11.2 Use of Artificial Intelligence as a Computational Tool

We employed four AI systems as computational instruments during the preparation of this work: ChatGPT (OpenAI), Claude (Anthropic), DeepSeek, and Manus. Their role was strictly analogous to that of a computer algebra system or proof assistant. We limited their use to:

- Symbolic manipulation of tensor equations and metric variations;
- Numerical evaluation and cross-checking of derived quantities;
- Literature cross-referencing and verification of mathematical identities;
- Translation of geometric insights into standard mathematical notation and $\text{\LaTeX}$ formatting.

We used no AI system to generate original physical insights, propose hypotheses, or derive results independently. All foundational ideas—including the identification of $\kappa = 3$, the exhausted-current limit, the holonomy deficit $\Delta = (\pi - 3)/\pi$, the scale ladder $L_n = \ell_P \times 3^n$, and all specific predictions—were formulated by the author prior to AI interaction. The author’s process involves recognizing structural and geometric patterns across disparate physical and mathematical domains. We then employ AI tools to formalize these observations into the conventional language of theoretical physics.

11.3 Verification and Reproducibility

We registered all predictions on viXra:2512.0067 (26 December 2025) and Zenodo (DOI: 10.5281/zenodo.19022053) prior to experimental confirmation. We computed numerical values using high-precision arithmetic. We cross-checked them against standard reference values from the Particle Data Group, DESI, SH0ES, and other collaborations. We reproduce all equations and derivations in full. We state the underlying assumptions explicitly.

12 Conclusion

The $\kappa = 3$ stability constant provides a unified geometric origin for both General Relativity and Quantum Mechanics. By replacing continuous spacetime with a discrete A_2 hexagonal lattice, we derive the Born rule, the Einstein field equations, the Bekenstein-Hawking entropy, and the PMNS mixing angles without continuous parameter fitting. We registered the framework's predictions prior to confirmation. Multiple falsifiable tests remain for near-term experiments.

Independent peer-reviewed work from Maldacena, Anastasiou & Olea, Ghilencea, Bender & Mannheim, Fabbri, Mannheim & O'Brien, Dechant, Boyle & Farnsworth, Cheung et al., and others provides rigorous mathematical and empirical support for every layer of the framework. The hexagon is the foundation.

Dedication

To my wife, Kirsty — who felt this long before it had a name.

Before there were notes, models, or explanations. There was her endurance, and the quiet cost of carrying something no one else could yet see.

This work exists because of you. Before I understood. Clarity comes from you. It was earned through what you bore first.

It was because of you.

Thank you, my darling.

Rusty — you put the R into my universe.

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The author is indebted to the foundational work of Albert Einstein. His insistence on geometric simplicity and causal strictness remains the guiding standard for physical theory. In the one hundred and tenth year since the publication of the field equations, it is a privilege to submit this geometric continuation.

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